Ying Xu

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EDUCATION

North China Electric Power University 2023.09 – 2027.06 (expected)
B.Eng. in New Energy Science and Engineering (wind-energy focus)

- GPA: 3.29/5.00; calculated weighted average: 82.9/100.
- Selected coursework: Wind Resources Measurement and Assessment (95/100), Numerical Simulation (90/100), Reliability Engineering (90/100), Wind Turbine Aerodynamics (Excellent).

RESEARCH INTERESTS

Wind-energy integration; wind-turbine modeling and dynamics; renewable power systems.

RESEARCH EXPERIENCE

Multi-Timescale Modeling of PMSG Wind Turbines 2026.09 – present
Undergraduate research project, NCEPU · Advisor: Dong Zhao

- Reviewed 12 papers on wind-turbine models, model reduction, voltage ride-through, and frequency response; developed an initial framework separating mechanical, power-control, and current-control dynamics.
- Current focus: model formulation and parameter specification for a direct-drive permanent-magnet synchronous generator (PMSG) turbine; planned evaluation covers wind-speed, voltage, and frequency disturbances.

CFD of Cylinder Vibration and Airfoil Dynamic Stall 2026.07 – 2026.08
Wind Power Research Center, NCEPU · Supervised by a doctoral researcher

- Independently completed five 2D Fluent case types under guidance: cylinder flow/vortex-induced vibration, NACA0012 static aerodynamics/dynamic stall, and S809 dynamic stall; ran cylinder flow in OpenFOAM.
- Analyzed cylinder vibration and force spectra; obtained lift/drag frequencies $fD/U_\infty \approx 0.166/0.334$ at $Re = 100$.
- Compared NACA0012/S809 load hysteresis with published experiments, identifying NACA0012 deviations near stall and during pitch-down, and underpredicted S809 pitch-down lift; prepared a technical presentation.

AWARDS & HONORS

- **National First Prize**, TI Cup National Undergraduate Electronic Design Contest 2025 (nominated for the TI Cup)
- National Third Prize, National Undergraduate Electronic Design Contest 2024 — Analog Electronic System Design Contest (TI Cup)
- First Prize, Beijing Undergraduate Electronics Design Contest (2024)
- University Merit Student; Second-Class University Scholarship; Science and Technology Innovation Scholarship

SELECTED ENGINEERING PROJECTS

WindAssess — Wind-Resource Assessment Toolkit 2026.06

- Built a Python pipeline for SODAR/LIDAR and met-mast wind data, with range, correlation, trend, and missing-data checks; implemented Weibull fitting, wind-shear estimation, air-density calculations, and wind roses, with unit tests.

Monocular-Vision Target Measurement Device 2025.07 – 2025.08
National First Prize team · Hardware and system integration

- Designed the power-monitoring circuit, adapted the MIPI display, and integrated hardware and software on an embedded Linux platform during a four-day competition.
- The team achieved target-dimension errors below 2 mm at approximately 1.6 m; measured system power was 7 W peak and 5 W idle, within the 10 W limit.

12 V Power Monitor 2026.09 – present

- Built a custom PCB and ESP32-C3 firmware for current/power measurement, with hardware and software overcurrent protection; developed a Python backend for data logging and live visualization.

NanoScope — Two-Channel Pocket Oscilloscope 2025.06 – 2025.07

- Built a dual-channel instrument with an analog front end, STM32 ADC/DMA acquisition, and ESP32 display; implemented frequency/phase measurement over 50 Hz–200 kHz and Bluetooth control.

LEADERSHIP

Head of Technology Department, MOS Innovation Lab, NCEPU 2025.05 – present

- Organized STM32 training and mentored new members; designed technical assessments and led project reviews for dozens of members.

TECHNICAL SKILLS

Simulation	ANSYS Fluent (2D external flow); OpenFOAM (basic); Gmsh; ParaView
Programming	Python, C; MATLAB (basic)
Engineering	STM32 / ESP32, PCB design, embedded Linux; Git, Docker, SolidWorks
Languages	English: IELTS Academic 7.0 (September 2026)